

NSCP 2015 CONCRETE STAIRWAY DESIGN REPORT

Project: Concrete Stairway Design
Date: October 18, 2025
Code: NSCP 2015
Designer: Structural Engineer

1. MATERIAL PROPERTIES

Property	Value	Unit
Concrete f'_c	20.7	MPa
Steel f_y	275.0	MPa
Concrete γ	24.0	kN/m ³
Beta (β ■)	0.85	-
ρ min	0.005091	-
ρ max	0.027969	-
ρ balanced	0.037292	-

2. GEOMETRY AND LOADS

Parameter	Value	Unit
Tread Width	0.25	m
Riser Height	0.175	m
Clear Span	3.0	m
Steps - First	8	steps
Steps - Second	8	steps

Load Type	Value	Unit
Live Load	1.9	kPa
Misc Live	0.5	kPa
Floor Finish	1.0	kPa
Misc Dead	0.5	kPa

Support Configuration for Design

First Support: Two-point support with stringers

Second Support: Two-point support with stringers

3.1 FIRST FLIGHT DESIGN RESULTS

Parameter	Value
Support	Two-point support with stringers
Min Thickness (mm)	150.0
Adopted Thickness (mm)	170.0
Effective Depth (mm)	144.0
Dead Load (kPa)	8.58
Live Load (kPa)	2.40
Factored Load (kPa)	14.14
Max Moment (kNm)	15.90
Steel Area (mm ²)	733.09
Steel Ratio	0.005091
Main Bar Spacing (mm)	154.3
Temp Steel Area (mm ²)	306.00
Temp Bar Spacing (mm)	256.7

3.2 SECOND FLIGHT DESIGN RESULTS

Parameter	Value
Support	Two-point support with stringers
Min Thickness (mm)	150.0
Adopted Thickness (mm)	170.0
Effective Depth (mm)	144.0
Dead Load (kPa)	8.58
Live Load (kPa)	2.40
Factored Load (kPa)	14.14
Max Moment (kNm)	15.90
Steel Area (mm ²)	733.09
Steel Ratio	0.005091
Main Bar Spacing (mm)	154.3
Temp Steel Area (mm ²)	306.00
Temp Bar Spacing (mm)	256.7

REINFORCEMENT SCHEDULE

First Flight:

- Main Reinforcement: 12 mm ■ @ 154 mm c/c
- Temperature Bars: 10 mm ■ @ 256 mm c/c

Second Flight:

- Main Reinforcement: 12 mm ■ @ 154 mm c/c
- Temperature Bars: 10 mm ■ @ 256 mm c/c

4. DESIGN SUMMARY

The concrete stairway has been designed in accordance with NSCP 2015 standards, considering all specified parameters and support configurations. All reinforcement ratios are within acceptable limits and the design satisfies strength and serviceability requirements.